

Types and Variables

Julia is a dynamically typed language, but it is possible to specify the type of a variable with the ‘::’ operator.

Lowlevel types

<code>Int64</code>	64-bit signed integer.	<code>UInt64</code>	64-bit unsigned integer.
<code>Int32</code>	32-bit signed integer.	<code>UInt32</code>	32-bit unsigned integer.
<code>Int16</code>	16-bit signed integer.	<code>UInt16</code>	16-bit unsigned integer.
<code>Int8</code>	8-bit signed integer.	<code>UInt8</code>	8-bit unsigned integer.
<code>Float64</code>	64-bit floating point.	<code>Float32</code>	32-bit floating point.
<code>Float16</code>	16-bit floating point.	<code>Float8</code>	8-bit floating point.
<code>bool</code>	Boolean type.	<code>Char</code>	Character type.

Mathematical types

The following types are abstract types, which means that they cannot be instantiated directly, but they can be used to define other

<code>isequal(x,y)</code>	Check if 2 different types variables are equal.
<code>abs(x)</code>	Get the absolute value of <code>x</code> .
<code>abs2(x)</code>	Get the absolute value of <code>x</code> squared.
<code>inf</code>	Positive infinity constant.

Reals

<code>Real{T}</code>	Real number with type <code>T</code> .
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Complexes

<code>Complex{T}</code>	Complex number with type <code>T</code> real and imaginary parts.
<code>im</code>	Imaginary unit, e.g., <code>1.0im</code> .
<code>imag(z)</code>	Get the imaginary part of the complex number <code>z</code> .
<code>real(z)</code>	Get the real part of the complex number <code>z</code> .
<code>conj(z)</code>	Get the complex conjugate of the complex number <code>z</code> .
<code>angle(z)</code>	Get the phaseangle of the complex number <code>z</code> .

Rationals

<code>Rational{T}</code>	Rational number with type <code>T</code> numerator and denominator.
<code>numerator(r)</code>	Get the numerator of the rational number <code>r</code> .
<code>denominator(r)</code>	Get the denominator of the rational number <code>r</code> .